

# CE 490 - Introduction to Digital Image Processing

## Project Proposal: Enhancing Degraded Grayscale Images

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### 1 Objective

The objective of this project is to design a multi stage image restoration process using a single algorithm capable of removing various types of artifacts such as salt and pepper noise, Gaussian noise, blurring, and loss of fine details from five different grayscale images. This process aims to optimize performance metrics including Mean Squared Error (MSE), Peak Signal to Noise Ratio (PSNR), and Structural Similarity Index (SSIM), and to produce the most optimal solution.

### 2 Investigation of the Degraded Images

The project initially began with direct attempts to restore the degraded images using standard enhancement techniques. These efforts resulted in limited improvement. Consequently, our focus shifted toward understanding how the images were degraded. First, Fast Fourier Transform of the images were investigated. As shown in Figure 1, the FFT comparison shows that the degraded image contains both blur and Gaussian noise. The weakened high-frequency components indicate loss of fine detail due to blurring, while the more uniformly scattered frequency content reflects the presence of additive Gaussian noise. Additionally, visual inspection in the spatial domain shows there is salt-and-pepper noise in the degraded images.

After analyzing the original and degraded images, the following steps are applied to the original image in order to get an artificial degradation: a Gaussian blur with a standard deviation of 1.75, additive Gaussian noise with a variance of 0.01, and salt-and-pepper noise with a proportion matching the observed black and white pixels in the degraded image. Comparing histograms of the real and artificial degradations, an example shown in Figure 2, showed a similarity with the mean pixel-wise absolute intensity difference between the real and synthetic degradations corresponded to approximately 3% pixel wise error, indicating the chosen parameters closely approximate the actual noise characteristics.

### 3 Literature Background

Classical image restoration pipelines frequently adopt a multiple staged design, where different types of degradations are handled sequentially. In such schemes allow each

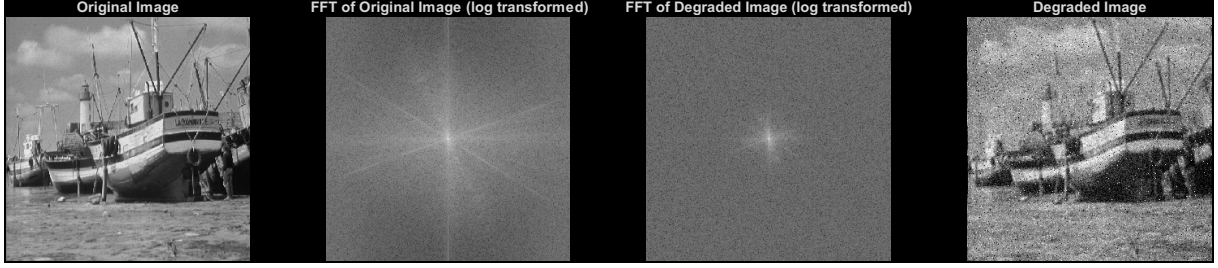


Figure 1: FFT analysis of the original and degraded image.

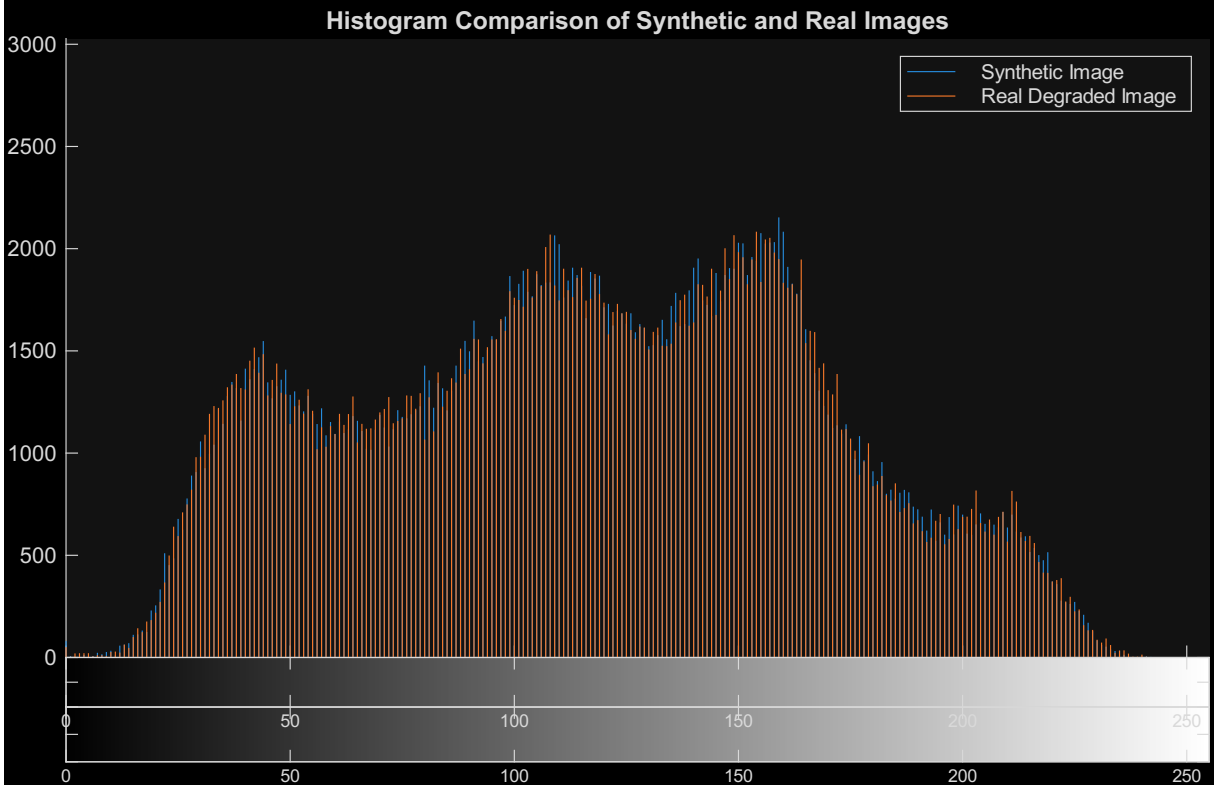


Figure 2: Histogram Comparison of Synthetic Degraded and Original Degraded Image

component to target a specific artefact. According to the finding in the previous section, the following subsections focus on methods for impulse noise suppression, Gaussian noise suppression, and detail enhancement.

Salt & pepper noise occurs when a fraction of pixels are randomly corrupted to the minimum or maximum intensity values. Median filters have been shown to be well suited for suppressing impulsive outliers [1]. However, standard median filters use a fixed window size and can over smooth details. Adaptive median filters solve this by adjusting the window size according to local statistics, which helps with well distinguish between noisy and noise-free pixels. It improves restoration under strong salt-and-pepper noise while preserving important structures [2, 3].

Classical linear smoothing operators, such as mean and Gaussian filters, are widely used to reduce Gaussian noise. However, because their weights depend on spatial distance, they tend to blur edges and fine structures together with the noise [1]. In order to cope with this limitation, edge-preserving smoothing is frequently used to handle Gaussian noise while maintaining significant edges. The bilateral filter, introduced by Tomasi and Manduchi [4], performs nonlinear averaging in both the spatial and intensity domains.

Pixels are weighted according to their spatial distance and intensity similarity. It allows homogeneous regions to be smoothed while strong edges are largely preserved. By using this technique instead of classical smoothing methods, there is an improvement in the restoration.

Enhancement of structural details and edges is another important component of restoration pipelines. Laplacian-based sharpening, and gradient operators have long been used to enhance edges, but these filters can easily amplify noise and introduce artefacts [1]. To reduce this effect, it is common to post process edge responses by thresholding and smoothing. For example, Canny’s classical edge detector combines Gaussian smoothing with gradient computation and thresholding in order to suppress noise and retain only significant edges [5].

## 4 Methods Planned

The planned methodology is based on a multiple staged restoration pipeline. This design should be considered as a preliminary that may be refined as the project progresses. Further literature review and different combination of enhancement techniques will be carried out to improve the approach. The current version of the pipeline is described below.

The restoration pipeline itself consists of three main stages. In the first stage, an adaptive median filter with a maximum window size of  $5 \times 5$  is applied to the degraded images in order to suppress salt & pepper noise while preserving as much structural detail as possible. After handling the salt & pepper noise, images yielded great PSNR values.

In the second stage, a bilateral filter is used to reduce noise and smooth regions while maintaining edges. The parameters of the bilateral filter are not chosen manually. Instead, a grid search is performed. For each parameter combination, the filtered output is compared against the original clean image using MSE, PSNR and SSIM, and the most optimal parameter combination is selected for that particular image. At the end, the bilateral filter is applied with these optimal parameters.

The third stage focuses on sharpening of structural details. First, a Laplacian filter is applied to the image to extract fine detail information, which is then thresholded using the first quartile (Q1) as the threshold value in order to suppress weak, noise-like edges. In parallel, a Sobel filter is applied to obtain edges and the resulted edges are normalized to  $[0, 1]$ , and scaled with a small weight to obtain a gentle edge contribution. These two components are combined to form an edge strength mask, which is further smoothed using a Gaussian filter. After retrieving the details, the extracted detail information is combined with the image from the second stage to obtain the sharpened result.

For each image, the restored result is quantitatively evaluated against the corresponding clean original using MSE, PSNR and SSIM. Finally, the metrics of the final stage are aggregated to compute global average MSE, PSNR and SSIM, yielding a global average MSE of approximately 312, a PSNR of about 24 dB, and an SSIM of 0.6.

## 5 Expected Outcomes

The expected outcome of this project is to develop an image restoration pipeline that is capable of enhancing degraded images that is implemented in MATLAB. The

pipeline aims to achieve reduction in both salt & pepper and Gaussian noise, while improving the sharpness and clarity of the images' detail. The outcomes of the developed pipeline will be evaluated by using common image quality metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Mean Squared Error (MSE) as well as visual comparisons. The outcomes of the pipeline will be optimized by trying different combinations of image enhancement techniques.

## 6 Evaluation Plan

As it was stated in the previous section, the evaluation is based on both qualitative and quantitative metrics. Quantitative evaluation will be based on the MSE, PSNR, and SSIM metrics computed between the original clean image and the restored output. Each stage of the pipeline will be measured and investigated independently to identify its contribution to the restoration process, as well as the final quantitative values. Also, the average of the results of the restored images' will be evaluated together at the end. The aim is to achieve minimizing the MSE as possible, to make SSIM above 0.8 and to increase the PSNR above 30 dB. In addition to these numerical measures, visual inspection of the restored images will be carried out. This qualitative inspection will be focused on the sharpness, visibility of fine details, and the noticeable artefacts.

## References

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